Observability Pitfalls, Shortcut Learning, and Entity-Level Benchmarking in Machine Learning for Enterprise Cybersecurity: An Empirical Study on the LANL Dataset

Autonomous AI Cybersecurity Research Subsystem & Project Team Department of Computer Science & Cybersecurity Engineering October 2026

Base Prev.	Precision
0.0534	0.0000
0.0534	0.0000
0.000158	0.0000
0.000158	0.0002
0.000158	0.0002

Keywords—Cybersecurity Telemetry, Intrusion Detection, Advanced Persistent Threats, Lateral Movement, Shortcut Learning, Class Imbalance, Downsampling Observability, LANL Dataset.

1. Introduction and Problem Formulation

Modern enterprise security operations face an overwhelming volume of machine-generated telemetry. Enterprise networks generate hundreds of millions of daily authentication logs, process executions, network flow records, and DNS lookups. Within this immense haystack, Advanced Persistent Threat (APT) actors operate with stealthy, low-and-slow tradecraft, utilizing living-off-the-land techniques and legitimate credentials to move laterally across enterprise workstations. Consequently, malicious actions represent less than 0.01% of total enterprise events.

In response, the academic literature has increasingly applied supervised and unsupervised machine learning (ML) models to enterprise logs. A recurring dilemma in the field is that proposed ML architectures consistently report benchmark accuracies exceeding 98% and ROC-AUC scores above 0.95 in published papers, yet commercial SOCs experience catastrophic false alert fatigue and failure to detect real-world intrusions. In this paper, we conduct a rigorous forensic and experimental evaluation using the Los Alamos National Laboratory (LANL) comprehensive multi-source cybersecurity dataset (Kent, 2015), uncovering the structural reasons behind this disconnect.

2. Related Work and Research Gap

The LANL dataset is widely recognized as the premier public benchmark for enterprise-scale intrusion detection. Early foundational work by Turcotte et al. (2017) applied Poisson Matrix Factorization to bipartite authentication graphs, demonstrating that lateral movement anomalies manifest as rare user-host transition edges. Bowers et al. (2014) and Neil et al. (2013) focused on sequential path traversal across network topologies, demonstrating that multi-hop lateral movement requires path-based tracking. In deep learning, Tuor et al. (2017) evaluated recurrent neural networks and autoencoders for insider threat detection on CERT and LANL data, reporting high detection rates on dense synthetic activity sequences.

However, foundational critiques by Sommer and Paxson (2010) and landmark methodological audits by Arp et al. (2022) have highlighted pervasive pitfalls in published cyber machine learning benchmarks, including temporal data leakage, synthetic label contamination, and extreme class imbalance distortion (the base-rate fallacy). Our research addresses a critical gap: existing literature has failed to mathematically model how standard streaming downsampling alters attack observability, and how diurnal features create spurious temporal shortcuts that invert model ranking.

3. Telemetry Architecture and Sampling Infrastructure

The LANL dataset records 58 days of enterprise activity across four core modalities: authentication (auth.txt.gz), process execution (proc.txt.gz), network flows (flows.txt.gz), and DNS queries (dns.txt.gz), alongside a separate ground-truth log (redteam.txt) containing 749 discrete compromise events. Processing 1.6 billion raw events requires downsampling. To ensure strict scientific validity, our sampling architecture implements Algorithm R streaming reservoir sampling across 120 discrete hourly strata (Days 9 to 13), processing 409.37 million raw lines to extract 120,000 events (1,000 events/hour quota per modality).

Strict temporal boundaries were enforced without overlap: Day 9 [691200, 756000) serves as the Training partition (18 hours, 216 windows); Days 10-12 [756000, 1036800) serve as the Validation partition (78 hours, 936 windows); and Day 13 [1036800, 1123200) was held out as a blind, untouched test partition. The sampler operated completely blind to redteam.txt, preventing any label-informed selection.

4. The Empirical Failure of Enterprise-Wide Window Benchmarks

Our baseline benchmark (Benchmark B) aggregated sampled telemetry into global 300-second enterprise windows. A clean Random Forest trained on Day 9 achieved training ROC-AUC = 1.0000, but completely collapsed on validation: ROC-AUC = 0.3938 and Average Precision (AP) = 0.0454 (against a baseline prevalence of 0.0534). Sub-0.50 ROC-AUC indicates an inverted ranking: the model systematically assigned lower threat probabilities to true attack windows than to benign windows.

Root-cause auditing identified two mathematical reasons for this failure:

1. Physical Sampling Obliteration: The underlying enterprise stream generates ~1.5 million events per hour. Uniform sampling at 1,000 events/hour yields an inclusion fraction of pi ≈ 0.000701 (1 in 1,426 events). Across 117 discrete attacks on Day 9, the binomial probability that zero attack events enter the sample is (1 - 0.000701)^117 = 92.1%. Audit verification confirmed that 0 / 117 attack events survived into the sample. The sampled telemetry in attack windows consisted of 100% benign background traffic.

2. Diurnal Shortcut Inversion: Because the sampled data was pure background noise, the model latched onto time-of-day features. On Day 9 (Train), 100% of attack events occurred during daytime working hours (10:00-18:00 UTC). On Days 10-12 (Validation), the adversary shifted operations, executing 66% of attacks during nocturnal off-hours. Consequently, the model learned that 'off-hours = safe', driving validation ROC-AUC below 0.50. When auth_off_hours was ablated, ROC-AUC shifted to 0.4986 (an exact random coin toss), proving the remaining 22 enterprise aggregate features contained zero predictive signal.

5. Benchmark C: Active Host-Window Candidate Benchmark

To resolve global enterprise dilution, we designed Benchmark C, shifting the prediction unit from an enterprise window to an active host-window instance (h, W_k). An instance is defined whenever an enterprise host h appears in window W_k as an initiator or recipient of telemetry. This yielded 19,349 instances in Train (6 positive, prevalence = 0.03101%) and 82,254 instances in Validation (13 positive, prevalence = 0.01580% / 158 ppm).

We engineered 11 clean behavioral features: total events, source events, destination events, modality breakdowns (auth, proc, flow, dns), distinct peer diversity, role flags (is_src, is_dst), and lag-1 temporal change (delta_events). Models were regularized (L1 Logistic Regression with C=0.1; shallow Random Forest with max_depth=3, min_samples_leaf=15) and fit strictly on Training data.

Table 1: Validation Performance Comparison across Benchmark Generations

Abstract—Machine learning (ML) models proposed for enterprise intrusion detection frequently demonstrate near-perfect academic performance (ROC-AUC > 0.95) but collapse when deployed in production Security Operations Centers (SOCs). This paper presents a forensic investigation into the causes of this generalization failure using the comprehensive multi-source cyber security telemetry dataset from Los Alamos National Laboratory (LANL). Through rigorous experimental auditing, we uncover three structural pitfalls in standard evaluation protocols: (1) synthetic data leakage, which produced artificial ROC-AUC = 1.0000 in early prototypes; (2) uniform streaming downsampling, which inadvertently acts as an obliteration filter for discrete advanced persistent threat (APT) events (yielding a 92.1% probability of dropping all attack events from the sample); and (3) diurnal shortcut learning, where classifiers latch onto time-of-day features (e.g. auth_off_hours), causing severe performance inversion (ROC-AUC = 0.3938) when adversary operational hours shift between training (daytime) and validation (nocturnal). To establish a defensible benchmark, we design and implement Benchmark C, framing detection at the active host-window entity level (h, W_k) across 101,603 active instances. We prove that removing diurnal shortcuts eliminates performance inversion, achieving positive discriminative capability (ROC-AUC = 0.5561, Average Precision = 0.000247 vs 0.000158 prevalence) and detecting 61.5% to 69.2% of true compromised host-windows. Finally, we analyze the operational alert burden under extreme enterprise class imbalance (158 parts per million) and synthesize guidelines for fair literature comparison.

6. Operational Alert Burden and Threshold Calibration

A central requirement for real-world deployment is assessing operational alert burden. Under an extreme base rate of 158 ppm (0.0158%), default thresholding (0.50) yields 740.77 alerts/hour for the candidate Random Forest (detecting 9 of 13 attack windows) and 555.83 alerts/hour for L1 Logistic Regression (detecting 8 of 13 attack windows). By auditing the decision threshold on validation data, we demonstrate that threshold calibration controls operational alerting: setting T = 0.58 for the candidate Random Forest curtails alert volume to 82.13 alerts/hour (6,406 alerts across 78.0 hours, with TP=1 and FP=6,405; FPR=0.0779, Specificity=92.21%), while setting T >= 0.585 suppresses all alerts. For the candidate L1 Logistic Regression at T = 0.58, alert rate is 418.53 alerts/hour with 7 true positives detected (53.85% recall) and precision of 0.000214. Feature weight analysis confirmed that delta_events (+1.1965) and role indicators (is_src: +1.0297, is_dst: +0.8584) are the primary drivers of true compromise detection.

7. Discussion and Guidelines for Literature Comparison

Our findings provide a critical lens for evaluating the cybersecurity ML literature. Direct numerical comparison across studies is scientifically invalid unless evaluation protocols are normalized: (1) Random train-test splits must be rejected in favor of chronological non-overlapping partitions; (2) Sampling fractions must be reported; (3) AP must be benchmarked against class prevalence (P/N); and (4) Diurnal shortcuts must be ablated. Our ablated candidate's AP of 0.000247 represents genuine above-chance detection against a 0.000158 base rate, and its 69.2% attack recall proves that entity-level modeling partially recovers signal obscured by global aggregation.

8. Conclusion

This research establishes that high reported accuracies in enterprise cybersecurity ML frequently mask data leakage, sampling destruction, or diurnal shortcuts. By rigorously freezing historical baselines, isolating test data, and transitioning to an active host-window benchmark, we successfully eliminated negative ranking inversion and established a transparent, reproducible benchmark for enterprise intrusion detection.

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